

Program 9: Organize kubernetes pods using labels and namespaces

Step 1: Create the namespace.yaml

```
apiVersion: v1
kind: Namespace
metadata:
  name: dev
```

```
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 2: Create the pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

```

apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80

apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80

```

Step 3: Apply both the yaml files

```

1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl apply -f namespace.yaml
namespace/dev created
namespace/prod created
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl apply -f pods.yaml
frontend-pod created
backend-pod created
frontend-pod created
backend-pod created

```

Step 4: Verify the pods

```

1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl get pods -n prod
NAME          READY   STATUS             RESTARTS   AGE
backend-pod    0/1     ContainerCreating   0           9s
frontend-pod   0/1     ContainerCreating   0           9s
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           14s
frontend-pod   1/1     Running   0           14s

```

Step 5: Filter the pods

```

1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           42s
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           52s

```

Step 6: Delete the resources.

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl delete pods --all -n dev
pod "backend-pod" deleted
pod "frontend-pod" deleted
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl delete pods --all -n prod
pod "backend-pod" deleted
pod "frontend-pod" deleted

1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl delete pods --all
pod "kubia" deleted
pod "nginx-deploy-6f9664446b-9fvk9" deleted
pod "nginx-deploy-6f9664446b-fj7ps" deleted
pod "nginx-deployment-6f9664446b-7bzg4" deleted
pod "nginx-deployment-6f9664446b-r695t" deleted
pod "nodejs-pod" deleted
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl delete svc --all
service "kubernetes" deleted
service "kubia" deleted
service "nginx-deployment" deleted
service "nginx-service" deleted
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program9$ kubectl delete deployments --all
deployment.apps "nginx-deploy" deleted
deployment.apps "nginx-deployment" deleted
```